B3.1 Gas exchange (Animal)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b3-1-animal.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Plant gas exchange (B3.1.7–10) is in Plant Systems. Image credits are on the last slide of the slides for this subtopic.

THE PROBLEM AND THE SURFACE

B3.1.1 Why every organism needs gas exchange

pp. 273–274

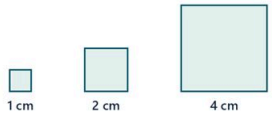
Cells respire continuously. Aerobic respiration consumes **oxygen** and produces **carbon dioxide**, and both gases move into or out of cells by diffusion. In a single-celled organism, the whole plasma membrane is enough to serve as the exchange surface.

Growth creates two problems. The **surface area-to-volume ratio** falls, so there is less surface for each unit of respiring tissue. The **distance** from the centre of the body to the outside also rises, and diffusion is only fast over very short distances. A tough, protective skin makes things worse.

So large, active animals have a specialized exchange organ, and usually a transport system as well.

Size makes gas exchange harder

B3.1.1



As an animal grows, its **surface area-to-volume ratio falls**: less surface for each unit of respiring tissue.

The **distance** from the centre to the outside also grows, and diffusion is only fast over tiny distances.

Cube side	Area	Volume	Area : volume
1 cm	6 cm^2	1 cm^3	6 : 1
2 cm	24 cm^2	8 cm^3	3 : 1
4 cm	96 cm^2	64 cm^3	1.5 : 1

Key idea
Large, active animals need a specialized exchange organ and a transport system.

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B3.1.2 What makes a good gas-exchange surface

p. 274

PROPERTY	WHY IT HELPS
PERMEABLE	Respiratory gases can actually cross it
THIN	Short diffusion distance
MOIST	Gases dissolve first, then diffuse across
LARGE SURFACE AREA	More gas can cross at once

B3.1.3 How animals keep concentration gradients going

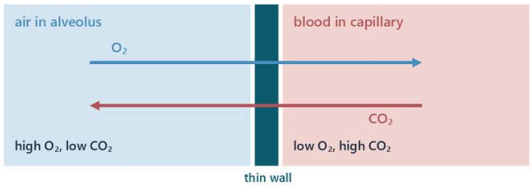
Net diffusion needs a concentration difference, and diffusion itself tends to remove that difference. Three features stop this from happening:

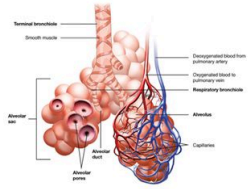
- **Dense networks of blood vessels** bring a large area of blood very close to the surface.
- **Continuous blood flow** carries oxygen away as soon as it arrives, and delivers more carbon dioxide.
- **Ventilation** replaces the medium outside: air for lungs, water for gills.

A fish must keep pumping water over its gills, because water that stops moving soon comes to equilibrium with the blood.

Keeping the gradient steep

B3.1.3





Blood vessels around alveoli. OpenStax College, CC BY 3.0.

Dense capillary network

A large area of blood lies close to the surface.

Continuous blood flow

Takes oxygen away and brings carbon dioxide.

Ventilation

Replaces the used air or water with fresh.

Diffusion evens out concentrations. Without these three, the gradient would vanish and exchange would stop.

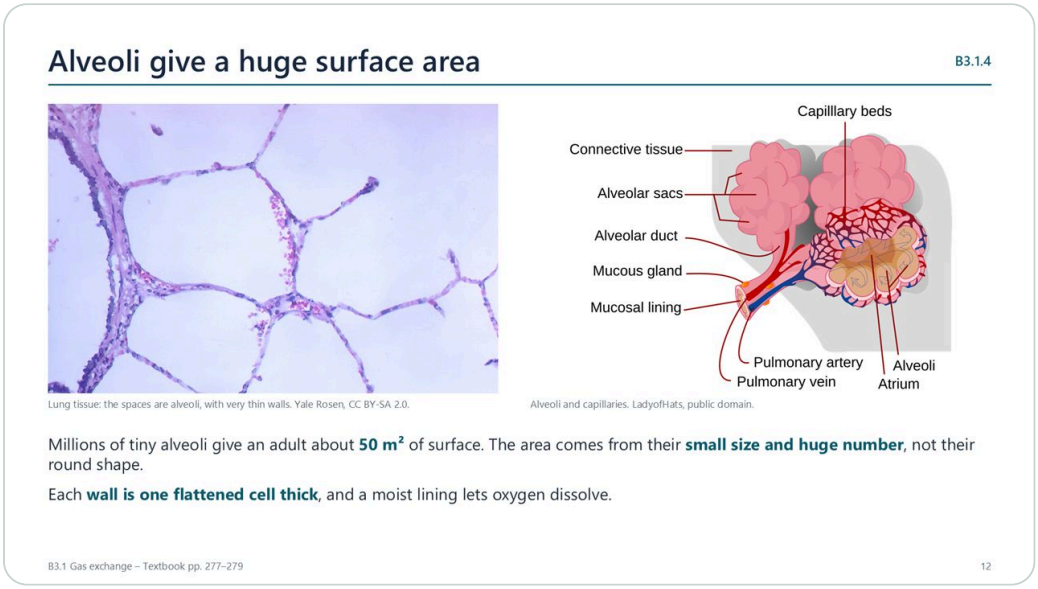
B3.1.4 How mammalian lungs are adapted to exchange gases

pp. 276–279

Air passes from the trachea to two bronchi, then through a tree of narrower and narrower **bronchioles**, ending in clusters of **alveoli**. Four adaptations are worth learning by name:

- **High surface area:** millions of tiny alveoli. The size and number give the area, not the round shape.
- **Branched bronchioles:** these distribute air evenly to every part of the lung.
- **Extensive capillary beds:** these add area and keep gradients steep. Blood low in oxygen arrives from the pulmonary artery and leaves through the pulmonary vein.
- **Surfactant:** a phospholipid-rich secretion that lowers the surface tension of the water lining. Without it, the moist walls would stick together and alveoli would collapse during exhalation.

The alveolar wall is one layer of flattened cells, so the diffusion path is short, and oxygen dissolves in the moist lining before it diffuses into the blood.



Watch out: the alveolar *wall* is one cell thick, not the alveolus. Do not confuse "alveolar wall" with "cell wall". Exhaled air still contains plenty of oxygen, so do not write that we breathe in oxygen and breathe out carbon dioxide.

B3.1.5 How the lungs are ventilated

pp. 279–281

The thorax is airtight, so changing its volume changes the pressure inside it. Air then flows from higher to lower pressure. Muscles can only pull, so opposite movements need **antagonistic** muscles.

	INHALATION	EXHALATION
DIAPHRAGM	Contracts, flattens and moves down	Relaxes, returns to a dome
ABDOMINAL MUSCLES	Relax	Contract, pushing the diaphragm up
EXTERNAL INTERCOSTALS	Contract	Relax
INTERNAL INTERCOSTALS	Relax	Contract
RIBCAGE	Up and out	Down and in
THORAX VOLUME	Increases	Decreases
PRESSURE	Falls below atmospheric	Rises above atmospheric
AIR FLOW	In	Out

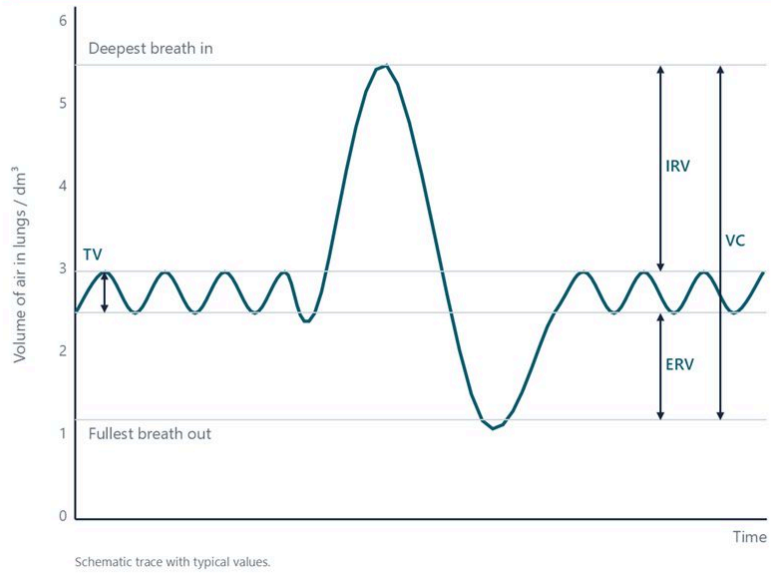
Watch out: keep the cause before the effect. Muscles change the volume, volume changes the pressure, and pressure moves the air. Say which intercostal muscles you mean every time.

B3.1.6 Measuring lung volumes

pp. 281–283

Reading a spirometer trace

B3.1.6



Schematic trace with typical values.

- **Tidal volume**: one normal breath, about 0.5 dm^3
- **Inspiratory reserve**: extra air in after a normal breath in
- **Expiratory reserve**: extra air out after a normal breath out
- **Vital capacity**: deepest breath in to fullest breath out

Equation

VC = TV + IRV + ERV

Here: 0.5 + 2.5 + 1.3 = 4.3 dm^3

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- **Tidal volume (TV):** the volume moved in one normal, relaxed breath, about 0.5 dm^3.
- **Inspiratory reserve (IRV):** the extra volume you can breathe in after a normal breath in.
- **Expiratory reserve (ERV):** the extra volume you can force out after a normal breath out.
- **Vital capacity (VC):** the largest volume you can breathe out after the deepest possible breath in. VC = TV + IRV + ERV.
- **Ventilation rate:** breaths per minute.

A spirometer gives a trace of volume over time. A lung volume bag, water displacement or a balloon give simpler estimates. Vital capacity varies with age, body size and fitness. During exercise, both tidal volume and ventilation rate rise.

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